

3(a)	(thermal) energy per unit mass (to cause temperature change)	B1
	(thermal) energy per unit <u>change</u> in temperature	B1
3(b)(i)	$(T =) pV / Nk$	B1
3(b)(ii)	$(pV =) NkT = \frac{1}{3}Nm\langle c^2 \rangle$ or $pV = NkT$ and $pV = \frac{1}{3}Nm\langle c^2 \rangle$	M1
	leading to $\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$ and $\frac{1}{2}m\langle c^2 \rangle = E_K$	A1
3(b)(iii)	internal energy = ΣE_K (of molecules) + ΣE_P (of molecules) or no forces between molecules	B1
	potential energy of molecules is zero	B1
3(c)(i)	increase in internal energy = Q + work done	B1
	constant volume so no work done	B1
3(c)(ii)	$c = Q / Nm\Delta T$	C1
	$= [N \times (3/2)k\Delta T] / (Nm\Delta T) = 3k / 2m$	A1
3(d)	(as it expands) gas does work (against the atmosphere/external pressure)	B1
	for same temperature rise) more (thermal) energy needed, so larger specific heat capacity	B1